Aishwarya Khedkar

+91 9890304043 | aishwaryank2025@gmail.com | Pune, India
GitHub: aishwaryaK2024 | LeetCode: aishwaryank24

Technical Skills

Languages	C, C++, Python
Core CS	Object-Oriented Programming, Data Structures & Algorithms
Technologies	CUDA, CMake, SSH
Tools	Git, VS Code, Jupyter, Canva, LaTeX
Libraries	NumPy, Pandas, OpenCV, Tkinter
Operating Systems	Windows, Linux (Ubuntu)

Education

M.Sc. in Scientific Computing **Pursuing**
Savitribai Phule Pune University, Pune

B.Sc. in Computer Science **2021 – 2024**
Savitribai Phule Pune University, Pune *CGPA: 7.9/10*

Projects

Numerical Computing Library **Jan 2026 – Dec 2026**
C++, CMake, OOP | Individual Project *GitHub*

- Designed and built a fully functional, open-source numerical computing library implementing Gaussian Elimination and LU Decomposition, structured entirely around Object-Oriented Programming principles.
- Applied core OOP concepts – class design, inheritance, polymorphism, and encapsulation – to keep the codebase modular, extensible, and maintainable.
- Built and distributed static and dynamic library variants using CMake, and version-controlled the project workflow with Git.

Zero Minor Detection (ECDLP) **Jul 2025 – Apr 2026**
C++, CUDA *Team Size: 4*

- Contributed to a computational cryptography research project solving the Zero Minor Problem, designing efficient algorithms to identify zero minors in large matrices.
- Applied data structures and algorithmic problem-solving to matrix computations relevant to the Elliptic Curve Discrete Logarithm Problem (ECDLP), collaborating within a 4-member team.

Sign Language Recognition System **Jan 2024 – Apr 2024**
Python, OpenCV, CVZone, Tkinter | Team Size: 2 *GitHub*

- Built a real-time computer vision application using OpenCV and CVZone hand-tracking to recognize and classify sign language gestures, debugging and iterating on detection accuracy.
- Integrated a Tkinter GUI to display live predictions, translating functional requirements into working, user-facing code.